

CME 307 / MS&E 311 / OIT 676: Optimization

Quadratic optimization

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Outline

Quadratic optimization

Quadratic approximations

Quadratic optimization

a **quadratic optimization** problem is written as

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T Qx + c^T x := f_0(x) \\ \text{variable} & x \in \mathbb{R}^n \end{array}$$

where

- ▶ $Q \in \mathbb{R}^{n \times n}$: symmetric positive semidefinite matrix
- ▶ $c \in \mathbb{R}^n$: vector

example: minimize least-squares objective

$$\frac{1}{2}\|Ax - b\|^2 = \frac{1}{2}x^T A^T A x - b^T A x + \frac{1}{2}\|b\|^2$$

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how to solve?

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how to solve? take gradient and set to 0:

$$\nabla f_0(x) = Qx + c = 0$$

$\implies$ linear system solvers also solve quadratic problems

Symmetric positive semidefinite matrices

Definition

a symmetric matrix $Q \in \mathbb{R}^{n \times n}$ is **positive semidefinite** (psd) if $x^T Q x \geq 0$ for all $x \in \mathbb{R}^n$.

these matrices are so important that there are many ways to write them! for $Q \in \mathbb{R}^{n \times n}$,

$$Q \in \mathbf{S}_+^n \iff Q \succeq 0 \iff Q = Q^T, \lambda_{\min}(Q) \geq 0$$

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why care about psd matrices Q ?

- ▶ least-squares objective has a psd $Q = A^T A$
- ▶ level sets of $x^T Q x$ are (bounded) ellipsoids
- ▶ the quadratic form $x^T Q x$ is a metric iff $Q \succ 0$
- ▶ eigenvalue decomp and svd coincide for psd matrices

Quadratic program

an equality constrained **quadratic program** is written as

$$\begin{array}{ll}\text{minimize} & \frac{1}{2}x^T Qx + c^T x \\ \text{subject to} & Ax = b \\ \text{variable} & x \in \mathbb{R}^n\end{array}$$

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how to solve? reduce to quadratic optimization problem:

- ▶ (explicit) form solution set $\{x : Ax = b\} = \{x_0 + Vz \mid z \in \mathbb{R}^{n-m}\}$ by computing a solution $Ax_0 = b$ and a basis V for the null space of A
- ▶ (implicit) use duality to recast problem as larger linear (KKT) system

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- ▶ (implicit) use duality to recast problem as larger linear (KKT) system
- ▶ inequality constraints: harder.

Solving equality-constrained quadratic program

$x^* \in \mathbb{R}^n$ solves the equality-constrained quadratic program

$$\begin{array}{ll}\text{minimize} & \frac{1}{2}x^T Qx + c^T x \\ \text{subject to} & Ax = b \\ \text{variable} & x \in \mathbb{R}^n\end{array}$$

$\iff$ there exists $\lambda^* \in \mathbb{R}^m$ such that

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \lambda^* \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$$

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proof: form Lagrangian

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^T Qx + c^T x + \lambda^T (Ax - b)$$

and solve for $\bar{x}$, $\bar{\lambda}$ so that $\nabla \mathcal{L}(\bar{x}, \bar{\lambda}) = 0$.

- ▶ $\frac{1}{2}\bar{x}^T Q\bar{x} + c^T \bar{x}$ provides an upper bound on p^* . (why?)
- ▶ $\frac{1}{2}\bar{x}^T Q\bar{x} + c^T \bar{x}$ provides a lower bound on p^* . (why?)

Quadratic program: application

Markowitz portfolio optimization problem:

$$\begin{array}{ll}\text{minimize} & \gamma x^T \Sigma x - \mu^T x \\ \text{subject to} & \sum_i x_i = 1 \\ & Ax = 0 \\ \text{variable} & x \in \mathbb{R}^n\end{array}$$

where

- ▶ $\Sigma \in \mathbb{R}^{n \times n}$: asset covariance matrix
- ▶ $\mu \in \mathbb{R}^n$: asset return vector
- ▶ $\gamma \in \mathbb{R}$: risk aversion parameter
- ▶ rows of $A \in \mathbb{R}^{m \times n}$ correspond to other portfolios
 - ▶ ensures new portfolio is independent, e.g., of market returns

Quadratic program: application

control system design problem:

$$x^+ = Ax + Bu$$

- ▶ $x \in \mathbb{R}^n$: state (e.g., position, velocity)
- ▶ $u \in \mathbb{R}^m$: control (e.g., force, torque)

$$\begin{aligned} & \text{minimize} && \sum_{t=1}^T x_t^T Q x_t + u_t^T R u_t \\ & \text{subject to} && x_{t+1} = Ax_t + Bu_t, \quad t = 0, \dots, T-1 \\ & && x_0 = x^{\text{init}} \end{aligned}$$

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Quadratic approximation

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable. For any $x \in \mathbb{R}$, approximate f about x :

$$f(y) \approx f(x) + \nabla f(x)^T (y - x) + \frac{1}{2} (y - x)^T \nabla^2 f(x) (y - x).$$

If f is a quadratic function, $\nabla^2 f(x) = H$ is constant.

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Quadratic approximations are useful because quadratics are easy to minimize:

$$\begin{aligned} y^* &= \underset{y}{\operatorname{argmin}} f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T H(y - x) \\ &\implies \nabla f(x) + H(y^* - x) = 0 \\ y^* &= x - H^{-1}(\nabla f(x)). \end{aligned}$$

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If we approximate the Hessian of f by $H = \frac{1}{t}I$ for some $t > 0$ and choose x^+ to minimize the quadratic approximation, we obtain the **gradient descent** update with step size t :

$$x^+ = x + -t \nabla f(x)$$

Quadratic upper bound

Definition (Smooth)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **L -smooth** if for all $x, y \in \mathbb{R}$,

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2.$$

Equivalently, assuming the derivatives exist,

- ▶ the operator $\frac{1}{L} \nabla f$ is **L -Lipschitz continuous**:

$$\|\nabla f(y) - \nabla f(x)\| \leq L \|y - x\|$$

- ▶ $\nabla^2 f(x) \preceq LI$ for all $x \in \text{dom } f$.

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A: $\lambda_{\max}(A)$ -smooth

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Q: For $A \succeq 0$, the quadratic function $f(x) = \frac{1}{2} x^T A x$ is ?-strongly convex

A: $\lambda_{\min}(A)$ -strongly convex

Some important functions

for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $x \in \mathbb{R}^n$,

- ▶ **Quadratic loss.** $\|Ax - b\|^2$
- ▶ **Logistic loss.** $f(x) = \sum_{i=1}^m \log(1 + \exp(b_i a_i^T x))$
where a_i is i th row of A

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Q: Which of these are strongly convex? Under what conditions?

A: Quadratic loss is strongly convex if A is rank n . Logistic loss is strongly convex on a compact domain if A is rank n .

Optimizing the upper bound

start at $x^{(0)}$. suppose f is L -smooth, so for all $y \in \mathbb{R}$,

$$f(y) \leq f(x^{(0)}) + \nabla f(x)^T (y - x^{(0)}) + \frac{L}{2} \|y - x^{(0)}\|^2$$

let's choose next iterate $x^{(1)}$ to minimize this upper bound:

$$\begin{aligned} x^{(1)} &= \operatorname{argmin}_y f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} \|y - x\|^2 \\ &\implies \nabla f(x^{(0)}) + L(x^{(1)} - x^{(0)}) = 0 \\ x^{(1)} &= x^{(0)} - \frac{1}{L} \nabla f(x^{(0)}) \end{aligned}$$

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- ▶ **gradient descent** update with step size $t = \frac{1}{L}$
- ▶ lower bound ensures true optimum can't be too far away...